THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle XYZ$, the side included between $\angle X$ and $\angle Y$ is: 1

(a) XY **(b)** XZ **(c)** YZ **(d)** none of these

2. Which set of angles can be the angles of a triangle? 1

(a) $30^\circ, 80^\circ, 80^\circ$ **(b)** $20^\circ, 80^\circ, 80^\circ$ **(c)** $20^\circ, 80^\circ, 90^\circ$ **(d)** $20^\circ, 70^\circ, 80^\circ$

3. Each angle of a right-angled isosceles triangle other than the right angle is: 1

(a) 90° **(b)** 60° **(c)** 45° **(d)** 30°

4. Two sides of a triangle are 7 cm and 15 cm. The third side could be: 1

(a) 12 cm **(b)** 24 cm **(c)** 22 cm **(d)** 8 cm

5. The three angles of a triangle are $(2x)^\circ$, $(3x)^\circ$ and 50°. If $x = 10$, the sum of the first two is: 1

(a) 50 **(b)** 52 **(c)** 51 **(d)** 49

6. The exterior angle of a triangle is $(3x + 15)^\circ$ and the interior opposite angles are 60° and 60°. The value of x is: 1

(a) 34 **(b)** 35 **(c)** 36 **(d)** 37

7. If $\triangle ABC \cong \triangle PQR$, then $\angle C$ equals: 1

(a) $\angle R$ **(b)** $\angle P$ **(c)** $\angle Q$ **(d)** $\angle A$

8. Which set of angles can be the angles of a triangle? 1

(a) $15^\circ, 75^\circ, 100^\circ$ **(b)** $25^\circ, 75^\circ, 90^\circ$ **(c)** $15^\circ, 65^\circ, 90^\circ$ **(d)** $15^\circ, 75^\circ, 90^\circ$

9. In an equilateral triangle the number of pairs of equal sides is: 1

(a) 6 **(b)** 2 **(c)** 3 **(d)** 1

10. The three angles of a triangle are $(4x)^\circ$, $(5x)^\circ$ and 30°. If $x = 8$, the sum of the first two is: 1

(a) 73 **(b)** 72 **(c)** 74 **(d)** 71

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, side BC is extended to D, $\angle ACD = (5x - 8)^\circ$, $\angle A = (2x + 12)^\circ$ and $\angle B = (x + 30)^\circ$. Find x and $\angle ACD$. 2

12. Two sides of a triangle are 5 cm and 5 cm. Find the least and the greatest whole-number length of the third side. 2

13. The angles of a triangle are $(2x + 8)^\circ$, $(3x - 10)^\circ$ and $(x + 8)^\circ$. Find x and the three angles. 2

14. In $\triangle ABC$, side BC is extended to D, $\angle ACD = (5x - 20)^\circ$, $\angle A = (2x + 8)^\circ$ and $\angle B = (x + 36)^\circ$. Find x and $\angle ACD$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 25)^\circ$ and $\angle B = (2x - 20)^\circ$. Find x and all three angles. 3

16. In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 9$ cm, $BC = QR = 11$ cm and $\angle B = \angle Q = 55^\circ$. Name the congruence rule, and find PR if $AC = 17$ cm. Find also the perimeter of $\triangle PQR$. 3

17. The exterior angle at C of $\triangle ABC$ is 110° and $\angle A : \angle B = 4 : 7$. Find all three angles of the triangle. 3

OR

In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 5)^\circ$ and $\angle B = (2x - 20)^\circ$. Find x and all three angles. 3

18. In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 7$ cm, $BC = QR = 8$ cm and $\angle B = \angle Q = 80^\circ$. Name the congruence rule, and find PR if $AC = 12$ cm. Find also the perimeter of $\triangle PQR$. 3

OR

The exterior angle at C of $\triangle ABC$ is 100° and $\angle A : \angle B = 1 : 4$. Find all three angles of the triangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The angles of a triangle are $(x + 35)^\circ$, $(x - 5)^\circ$ and $(2x + 6)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

OR

Two sides of a triangle are 10 cm and 13 cm. 5
(i) Between which two lengths does the third side lie?
(ii) How many whole-number lengths are possible?
(iii) Find the smallest and the largest possible perimeter in whole centimetres.

20. The angles of a triangle are $(x + 22)^\circ$, $(x - 12)^\circ$ and $(2x + 6)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

OR

Two sides of a triangle are 9 cm and 4 cm. 5
(i) Between which two lengths does the third side lie?
(ii) How many whole-number lengths are possible?
(iii) Find the smallest and the largest possible perimeter in whole centimetres.

END OF QUESTION PAPER — **Total: 40 Marks**